Storms of High-energy Particles

João Tomás and Tiago Viola, *Universidade Nova De Lisboa, Faculdade de Ciências e Tecnologia*

Abstract—In this paper we will explore how a benefectial a CUDA implementation on a sequential program can be.

Index Terms—Computer Science, Profiling, CUDA.

1 Introduction

In this paper we will explore how beneficial a CUDA implementation can be, to do this we first needed to get a sequential implementation of some heavy work, the one we used to this work was the [**?**]. This program simulates the effects of high-energy particles bombardment on an exposed surface, for example in the surface of space vessels in outer space.

mds
October 22, 2023

2 Methodology

Working methodology

2.1 Code Factorization

The initial phase involves a comprehensive analysis of the code to gain a deep understanding of its structure and functionality. Following this, a code factorization process is employed to enhance code organization and maintainability, which primarily involves reorganizing the code into well-defined functions.

2.2 Profiling

Once the code is refactored and modularized, the next step focuses on performance optimization. A profiler is utilized to identify performance bottlenecks within the codebase. The profiler helps pinpoint specific areas where improvements can be made to enhance the execution speed and resource efficiency.

2.3 CUDA implementation

With a clear understanding of the code's performance characteristics, the project moves forward to implement CUDA, a parallel computing framework. This CUDA implementation aims to leverage the power of GPUs to accelerate the code's execution. From now on our focus move to the parallelization of the more intense tasks that we were able to catch with the profiler, every time we did a change parallelized a function we run tests to compare to the sequential version.

- *M. Shell was with the Department of Electrical and Computer Engineering, Georgia Institute of Technology, Atlanta, GA, 30332. E-mail: see http://www.michaelshell.org/contact.html*
- *J. Doe and J. Doe are with Anonymous University.*

Manuscript received April 19, 2005; revised August 26, 2015.

3 Results

3.1 Profiler

Data from the first profiler and execution time from some of the inputs

3.2 Paralellization

Data from the second profiler and execution time for the same inputs of above.

4 Conclusion

Throughout the course of this project, we embarked on a journey of code optimization and parallelization that opened our eyes to the immense potential of modern tools and techniques. The utilization of profiling tools, some of which were entirely new to us, revolutionized our approach to performance optimization. These tools quickly proved to be indispensable, providing invaluable insights into our code's behavior, and allowing us to pinpoint and tackle performance bottlenecks with precision. What emerged as a remarkable revelation in this project was the exceptional speed and efficiency that a CUDA implementation can bring to computationally intensive workloads. The realization that a well-implemented CUDA solution can result in significant performance gains has reshaped our perspective on optimizing programs with heavy workloads.

Appendix A
Proof of the First Zonklar Equation

Appendix one text goes here.

Appendix B

Appendix two text goes here.

Acknowledgments

The authors would like to thank... FCT?